

Your grade: 100%

Next item →

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

1.

Let  $A = \{3^{-n} : n \in \mathbb{Z} \text{ and } 0 \leq n < 5\}$ . Which one of the following sets represents A using the listing method?

1 / 1 point

☐

$\{3, 27, 81\}$

☒

$\{1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \frac{1}{81}\}$

☐

$\{3, 9, 27, 81\}$

☐

$\{1, 3, 27, 81\}$

👍 Nice work

The possible values of n are 0, 1, 2, 3 and 4 hence, the elements of A are  $3^0, 3^{-1}, 3^{-2}, 3^{-3}, 3^{-4} = 1, 1/3, 1/9, 1/27, 1, 1/81$

2.

Let  $S = \{n : n \in \mathbb{Z}^+ \text{ and } n \bmod 2 = 0 \text{ and } n < 10\}$ . Which one of the following sets represents S using the listing method?

1 / 1 point

☐

$\{2, 4, 6, 8, 10\}$

☒

$\{2, 4, 6, 8\}$

☐

$\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$

☐

$\{0, 2, 4, 6, 8, 10\}$

👍 Nice work

S is the set of positive even integers less than 10

3.

Given the set  $S = \{x : x \text{ is a positive integer less than 10 and } x \bmod 2 = 0\}$ , which one of the following sets represents S?

1 / 1 point

☐

$\{0, 2, 4, 6, 8, 10\}$

☒

$\{2, 4, 6, 8\}$

☐

$\{0, 2, 4, 6, 8\}$

☐

$\{2, 4, 6, 8, 10\}$

👍 Nice work

S is the set of positive even integers less than 10, hence  $S = \{2, 4, 6, 8\}$

4.

Given the set  $S = \{n : n \in \mathbb{Z} \text{ and } 0 \leq n < 10 \text{ and } n \bmod 2 = 1\}$ . Which one of the following sets represents S?

1 / 1 point

☐

$\{0, 1, 3, 5, 7, 9\}$

☐

$\{1, 3, 5, 7, 9, 11\}$

☐

$\{0, 2, 4, 6, 8\}$

☒

$\{1, 3, 5, 7, 9\}$

👍 Nice work

S is just the set of positive odd integers less than 10